Control of Ventilation and Effects of Altitude and Exercise

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Adapted from: Control of Breathing

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Learning Objectives (6) Versions

After completing this brick, you will be able to:

1. Identify the regions in the central nervous system involved in control of breathing.
2. Describe the locations, roles of chemoreceptors and their importance in detecting changes in arterial partial pressure of oxygen (PO_2), carbon dioxide (PCO_2), and pH.
3. Describe pulmonary receptor reflexes that influence breathing frequency and tidal volume.
4. Describe the hypoxic respiratory drive in a COPD patient and changes with supplemental oxygen.

5. Describe pathological breathing patterns (Kussmaul, Cheyne-Stokes, Biot, and apneustic).

6. Explain adaptive ventilation changes with altitudes, exercise, and diving

CASE CONNECTION

GB is brought to the emergency department with an altered mental status and a history of illicit drug use. She is lethargic, with a decreased respiratory rate characterized by irregular breathing with episodes of apnea (absent respirations).

Given GB's presentation, why is she breathing abnormally and deeply? Consider your answer as you read, and we'll revisit GB at the end of the brick.

GO TO CONCLUSION

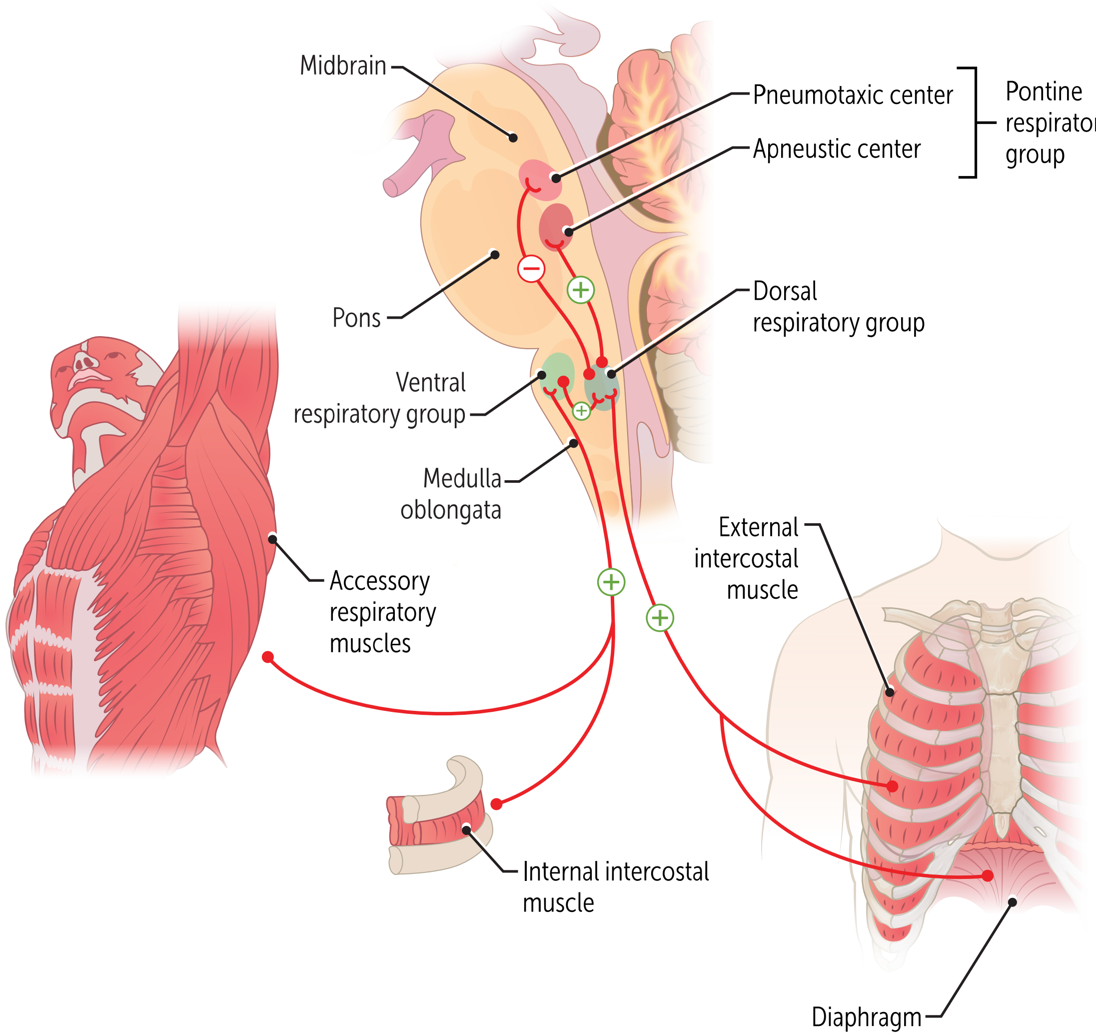
Why Is Breath Control Important?

Breathing is mostly an involuntary response, which is important since we cannot focus on it all the time. Breathing occurs when the respiratory muscles (eg, diaphragm, intercostal, and abdominal wall muscles) receive signals that originate in the brainstem (pons and medulla). These signals are controlled by input from different receptors: arterial chemoreceptors that sense blood gases, central chemoreceptors in the brain, stretch receptors in the lungs, and irritant receptors in the airways. Although breathing is automatic, we can also control it voluntarily. People can learn to control their breathing in activities such as swimming and singing, and in yoga where breath control is used for stress reduction

Neurologic inputs from the pons and medulla stimulate the respiratory muscles to initiate breathing.

How Does the Central Nervous System Control Breathing? (LO1)

The brainstem contains groups of neurons that regulate breathing by sending signals to the respiratory muscles. The **medulla** is the main control center, while the **pons** fine-tunes breathing patterns (Figure 1).



QUIZ

Tap image for quiz

Figure 1

Brainstem control of breathing (LO1)

Medulla: Primary Control Center

The **medulla**, found in the most inferior part of the brainstem, is the primary respiratory control center. The dorsal and ventral respiratory groups of neurons, both important for breathing, reside here.

Dorsal Respiratory Group (DRG). Located near the solitary nucleus, the DRG controls inspiration and sets the basic rhythm of quiet breathing. **Output:** efferent DRG sends signals through the phrenic and intercostal nerves to contract the diaphragm and external intercostal muscles to increase or decrease the respiratory rate. **Input:** The DRG receives afferent signals from the vagus (CN X) and glossopharyngeal (CN IX) nerves, which carry information from peripheral chemoreceptors in the blood and lung mechanoreceptors.

Ventral Respiratory Group (VRG). Usually inactive during quiet breathing. Becomes active during exercise or illness to drive forced inspiration and expiration.

The dorsal respiratory group generates the neural signals for normal quiet inspiration.

Pons– Pontine respiratory group: Modulatory center (LO1)

The **pons** helps shape breathing but is not essential for life. It contains two centers:

- The **pneumotaxic center (PNC)** acts as on "off switch" to inhibit inspiration by lowering tidal volume and respiratory rate
- The **apneustic center (APC**) stimulates inspiration and can cause long, gasping inspirations followed by brief, insufficient expirations (apneusis).
- *While this viewpoint is common in the literature, and likely to persist on Step 1 exams, clinically these are of historical significance (Richerson et al., 2017).*

Cerebral Cortex: Voluntary Control (LO1)

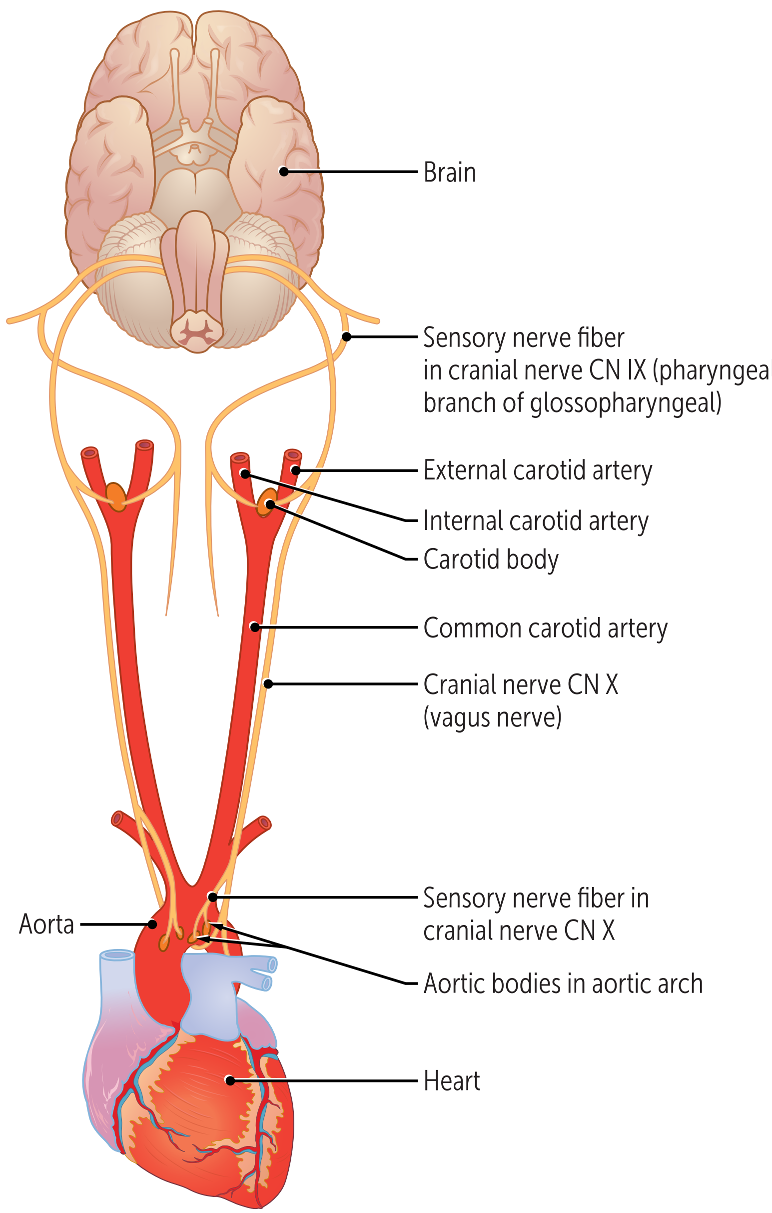
Your breathing is also under voluntary control, and can be modified for talking, swallowing, singing, defecation, coughing, and voluntary hypo- and hyperventilation.

What Chemical Sensors Control Breathing? (LO2)

The medulla and pons regulate ventilation based on input from chemoreceptors, which detect chemical changes, and mechanoreceptors, which detect physical changes in the lungs and airways. Here we focus on chemoreceptors, which sense changes in oxygen, carbon dioxide, and pH and then signal the medulla to alter breathing. Two main types exist: **peripheral chemoreceptors** in the arteries and **central chemoreceptors** in the brain.

Peripheral Chemoreceptors– primarily P_aO_2

Peripheral chemoreceptors, located in the peripheral arteries in carotid and aortic bodies (Figure 2), primarily respond to low arterial oxygen (**P_aO_2 < 75 mmHg**). They also respond to increased P_aCO_2 and low blood pH (acidemia, high H^+). Of these three signals, the **decreased P_aO_2 is primary** and most important.

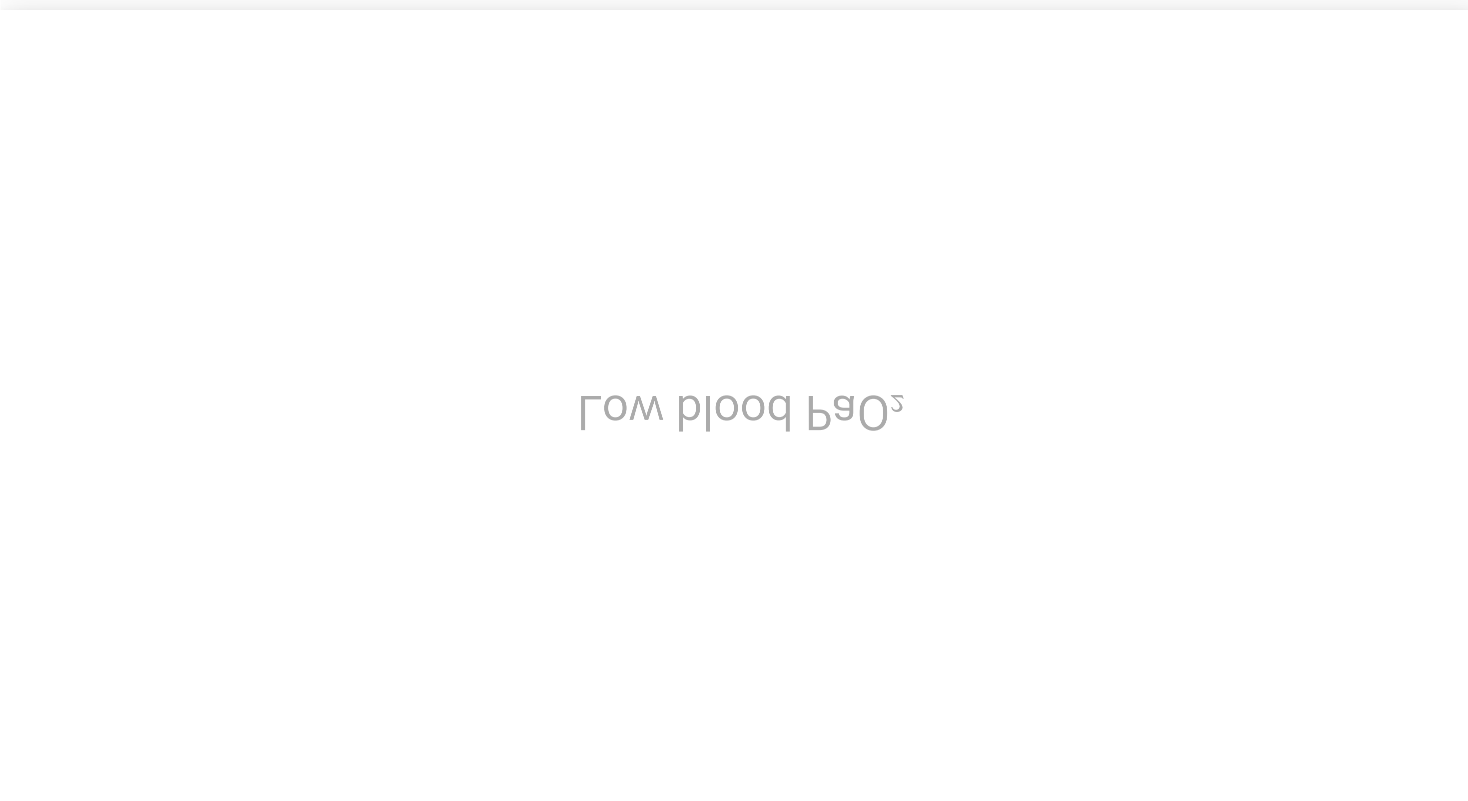


QUIZ

Tap image for quiz

Figure 2

These signals are carried to the medulla via the glossopharyngeal nerve (CN IX, carotid body) and the vagus nerve (CN X, aortic body). The medulla then stimulates the respiratory muscles to increase ventilation, raising P_aO_2, lowering P_aCO_2, and restoring pH.



Central Chemoreceptors– primarily CSF pH

Central chemoreceptors, located in the ventral medulla, are most sensitive to **decreased pH of the cerebrospinal fluid** (**CSF**). Increased blood P_aCO_2 will lead to more CO_2 crossing the blood-brain barrier into the cerebrospinal fluid (CSF). In turn, this increased dissolved CO_2 in the CSF increases carbonic acid, and in turn increases the hydrogen ion concentration in the

CSF (low pH). This decreased (acidic) pH causes the central chemoreceptors to increase the respiratory rate and tidal volume. Importantly, these chemoreceptors do not respond to low blood pH directly, since H^+ ions cannot cross the blood-brain barrier.

Decreased pH in the cerebrospinal fluid (CSF) due to increased $PaCO_2$. (Note: the trigger is NOT in response to low blood pH, as the H_+ ions cannot cross the blood brain barrier)

Ventilatory response to PO_2 and PCO_2 (LO2)

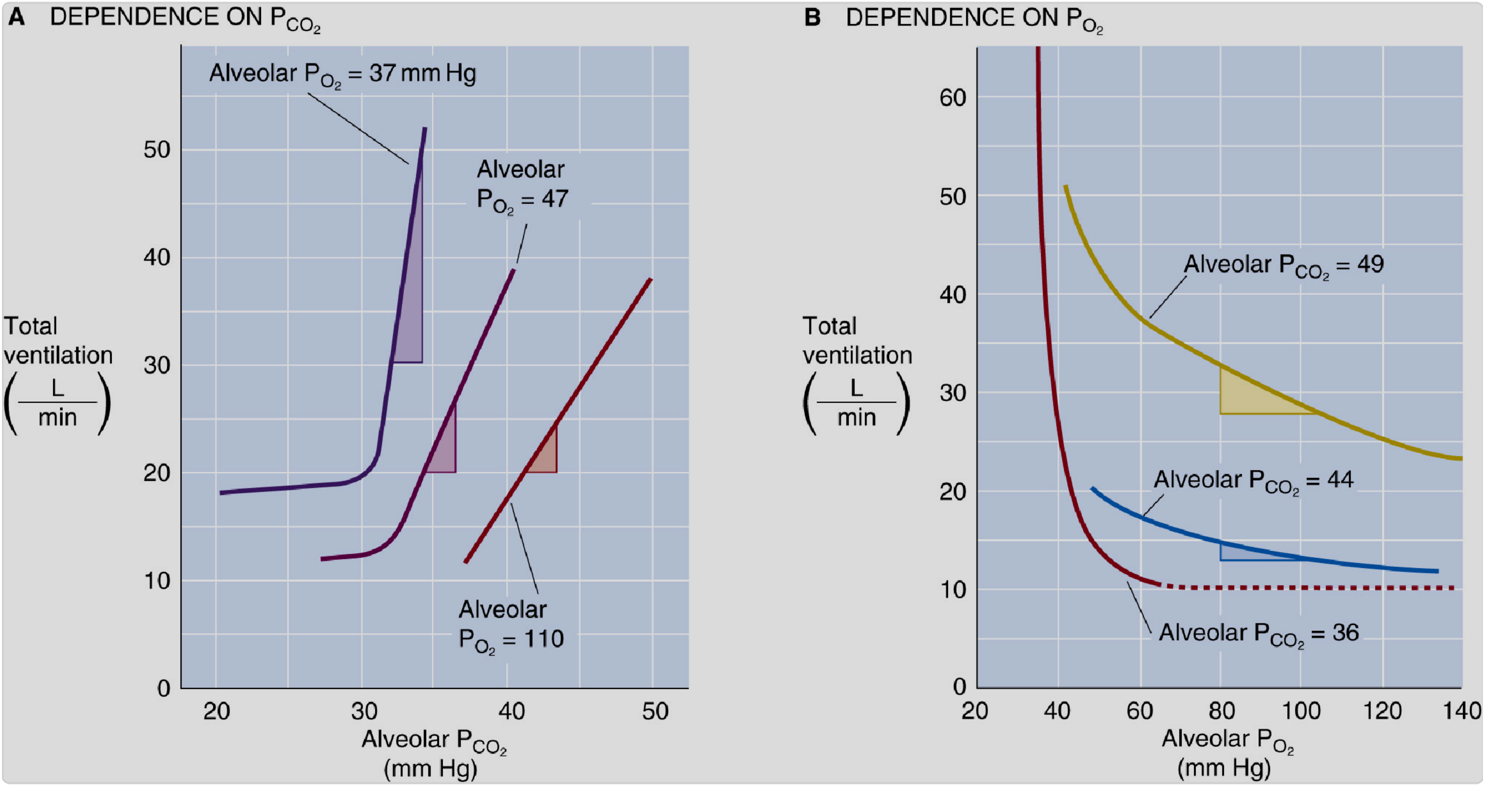


Figure 3. Ventilatory response to oxygen and carbon dioxide

Figure 3 shows how your body controls breathing by responding to two main signals: the amount of carbon dioxide (CO_2) and oxygen (O_2) in the lungs. **CO_2 is the main driver of ventilation**, but **low O_2 (hypoxemia) amplifies the body's response to CO_2.** Together, they work as a safety system: if CO_2 builds up or O_2 drops, the brainstem increases ventilation to restore balance.

Figure 3A shows that as **CO_2 increases,** ventilation (the amount of air you breathe per minute) also increases. Too much CO_2 in the blood is dangerous, so your body responds by breathing faster and deeper to "blow it off." This effect is even stronger when oxygen levels are low. In other words, hypoxia (low O_2) makes the body much more sensitive to CO_2.

Figure 3B shows **that as O_2 drops,** ventilation increases, particularly when the PO_2 starts to drop below 60 mmHg. This is a protective response to bring in more oxygen. The curves also show that this effect depends on how much CO_2 is present—if CO_2 is high, the drive to breathe at low O_2 becomes much stronger.

A low PaO_2 is detected in peripheral chemoreceptors, which signal the medulla to increase minute ventilation.

CLINICAL CORRELATION

Respiratory Drive with chronic hypercapnia e.g. COPD Patients (LO4):

In healthy people, breathing is mainly driven by CO_2 levels detected by central chemoreceptors. However, in chronic hypercapnia, as seen in **COPD** or emphysema, the brain's response to high CO_2 becomes blunted. These patients then rely more on low oxygen (**hypoxemia**) sensed by peripheral chemoreceptors to drive breathing.

Clinically, this shift is important. Giving too much supplemental oxygen can suppress the hypoxic drive, causing CO_2 retention and, in severe cases, **CO_2 narcosis** and respiratory failure. **Do you still give supplemental oxygen to these patients? Yes!** Instead, therapy is carefully titrated to keep oxygen saturation between **88–92%**, with close monitoring of blood gases.

What Mechanical Sensors Control Breathing? (LO3)

While chemoreceptors are the main drivers of breathing, **mechanoreceptors** also influence ventilation by detecting pressure and stretch in the lungs and airways (Figure 4). These include **pulmonary stretch receptors, irritant receptors, and J receptors**, with additional input from muscle and joint receptors during exercise. These mechanical receptors are summarizes in Figure 5 and Table 1.

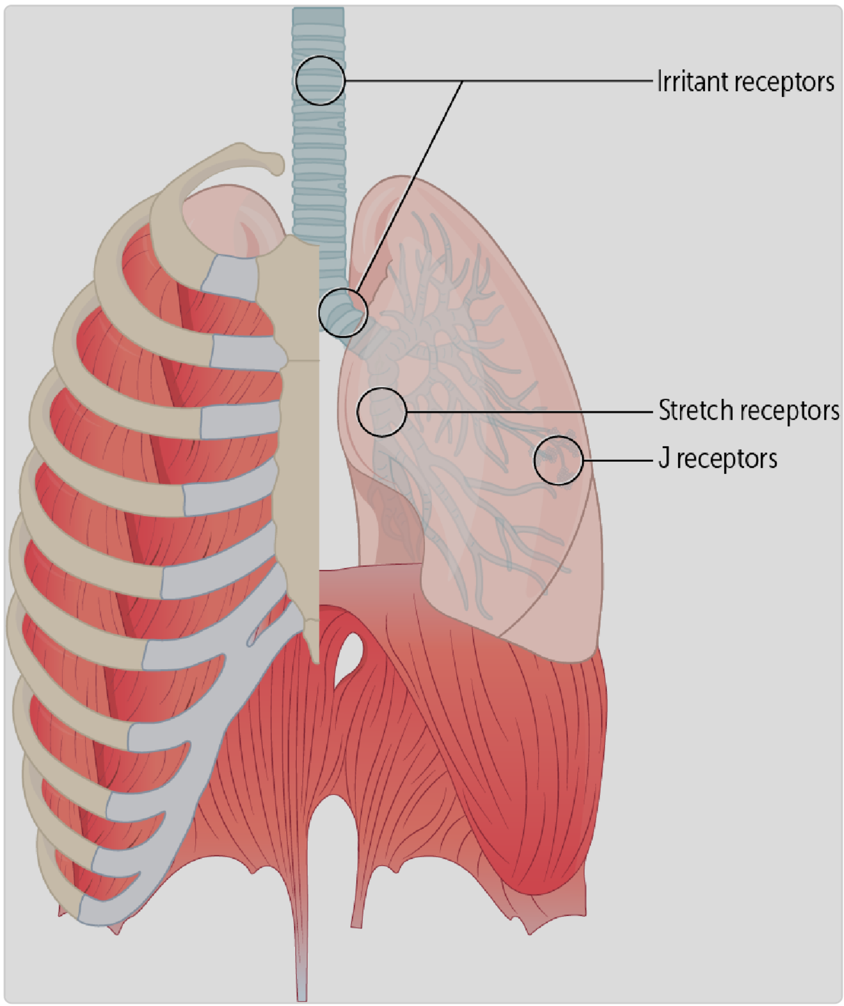


Figure 4. Mechanical sensors in the lung

Pulmonary Stretch Receptors

Pulmonary stretch receptors are located in airway smooth muscle and are activated when the **lungs expand (distend) during inspiration**. They send signals via the vagus nerve to the medulla, inhibiting further inspiration and preventing overinflation—a reflex called the **Hering-Breuer reflex**. This reflex does not usually affect normal breathing volumes but becomes important in diseases like **COPD**, where chronic lung overdistension weakens the reflex and slows breathing.

Irritant Receptors

Irritant receptors (aka rapidly adapting receptors), found in the airway epithelium of the trachea and bronchi, respond to **mechanical and chemical** irritants such as smoke, dust, or cold air. Afferent signals to the brain initiate coughing and bronchoconstriction, protective responses that clear harmful substances from the airways. This is one of the reasons why coughing comes naturally after exposure to fumes or smoke.

Juxta-pulmonary capillary receptors (J Receptors)

Juxta-pulmonary capillary receptors (J receptors) are unmyelinated C-fibers in the alveolar walls near capillaries. They respond to **increased pulmonary capillary pressure,** producing rapid, shallow breathing. This response is especially important in conditions like **heart failure**, where blood backs up into the lungs. J receptors also respond in conditions such as pulmonary edema and pulmonary emboli.

Mechanoreceptors- Joint and muscle receptors

Muscle and joint receptors contribute to the increase in breathing during exercise. They act as a **feedforward system**, signaling the brainstem to increase ventilation as soon as activity begins. (They are labeled as metaboreceptors in Fig 5).

Table 1. Summary of mechanical receptors that control breathing.

Receptor type	What they sense	Effect(s)
Pulmonary stretch receptors	Lung inflation	Inflation terminates
Irritant receptors	Noxious stimuli, eg, chemicals, dust, cold air	Coughing, bronchoconstriction
J receptors	Stretch (eg, pulmonary edema)	Shallow breathing
Mechanoreceptors	Joint and muscle stretch	Feedforward signals to increase breathing early in exercise

CLINICAL CORRELATION

The respiratory system uses certain behaviors to maintain gas exchange and prevent alveolar collapse (**atelectasis**) (Richerson et al., 2017).

Sighing briefly increases lung inflation, helping reopen collapsed alveoli and stimulating surfactant release. More frequent sighing during hypoxia or respiratory acidosis suggests it helps restore ventilation.

Yawning expands the lungs to full capacity for several seconds and is even more effective than sighing at reopening alveoli. Yawning often prevents atelectasis before sleep, reverses it after waking, and may also improve gas exchange before intense physical activity.

The figure below summarizes the role of the CNS and mechanoreptors in breathing (Figure 5).

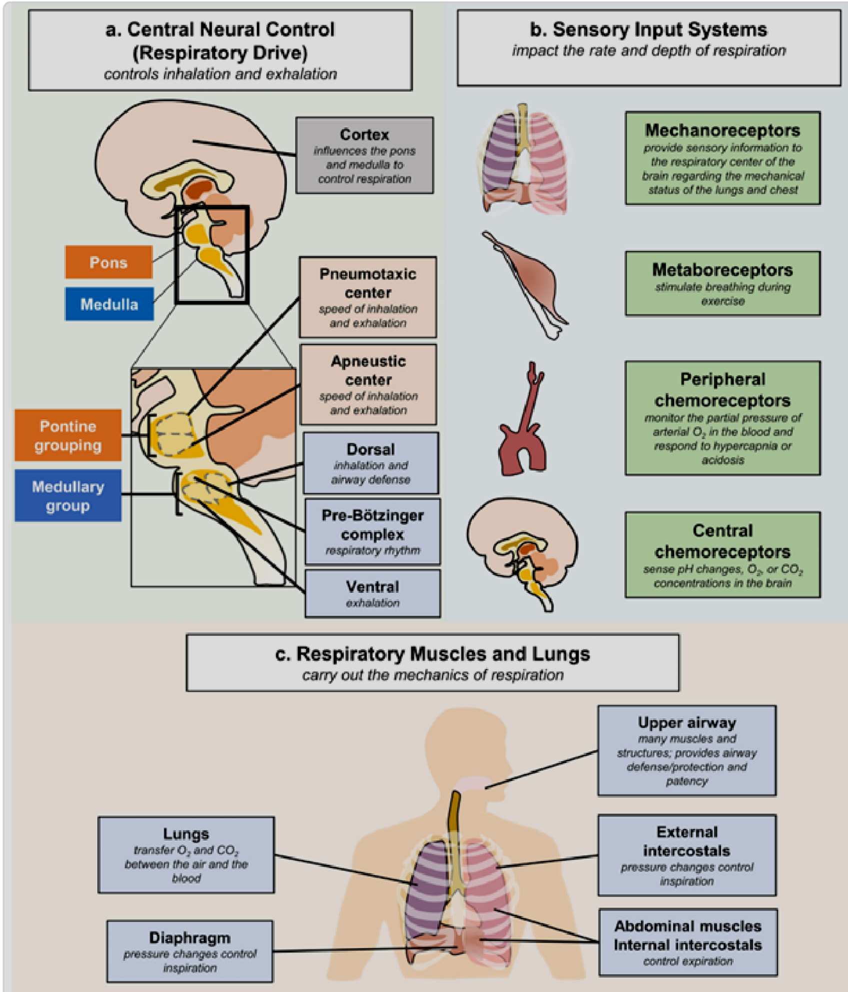


Figure 5. Summary of control of breathing

Pulmonary stretch receptors send impulses through the vagus nerve to inhibit inspiration.

What Are Some Pathological Breathing Patterns (LO5)?

Multiple sensors and central nervous system (CNS) regions are involved in breathing control, so there are also multiple opportunities for things to go awry. Let's compare a few abnormal breathing patterns (where the rhythm changes) with a normal pattern (Figure 6).

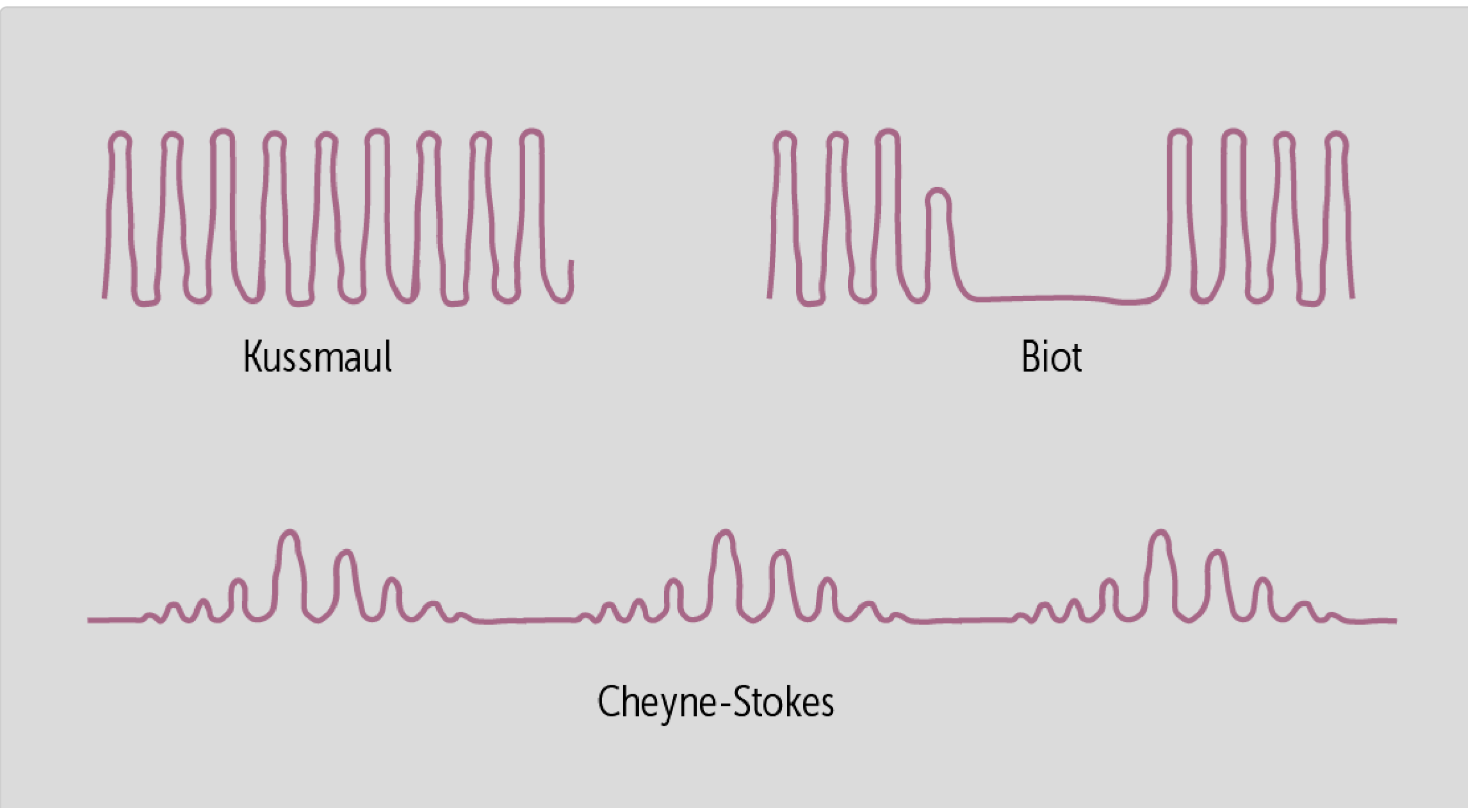


Figure 6. Patterns of breathing

These abnormal breathing patterns can be characteristic of certain disease or neurological states:

- **Kussmaul respirations** are deep, labored respirations, without pauses associated with metabolic acidosis (low blood pH due to an abnormally high levels of anions in blood). It is most often associated with diabetic ketoacidosis, a complication of type 1 diabetes mellitus in which the blood pH falls to low levels due to the formation of ketoacids.
- **Biot breathing** (also called **ataxic breathing**) is an irregular breathing pattern with increasing episodes of unpredictable apnea (absent respirations). It is most often associated with opioid intoxication or with injuries to the medulla.
- **Cheyne-Stokes respirations** are a characteristic pattern of alternating fast and slow breathing with periods of central apnea. With central apnea, the brain does not create the normal stimulation to breathe. It can be seen in patients with cardiac disease, neurologic disease, use of sedative drugs, acid-base disorders, and high-altitude exposure.
- **Apneustic breathing** are periods of prolonged inspiratory hold followed by short expirations due to lesions in the caudal pons affecting the apneustic center.

Biot breathing (or ataxic breathing) is often associated with injuries to the medulla.

CLINICAL CORRELATION

Sleep apnea is a group of disorders where breathing repeatedly stops during sleep, especially during REM. In **central sleep apnea**, the problem is reduced respiratory drive from the brain. More commonly, **obstructive sleep apnea** (OSA) occurs when the airway collapses during sleep due to decreased airway muscle tone and structural narrowing (Richerson et al., 2017). Obesity is a major risk factor for OSA, but enlarged tonsils, small jaw size, or other anatomical features can also contribute in non-obese patients. Symptoms include loud snoring, poor sleep quality, daytime sleepiness, and behavioral changes. Untreated cases may lead to **hypertension** and **cardiac arrhythmias**

Integrated responses of the respiratory system: high altitudes, exercise, and diving (LO6).

Respiratory responses to High Altitudes

At high altitudes, the partial pressure of oxygen (PO_2) decreases, leading to decreased alveolar oxygen (P_AO_2), and **hypoxemia** (low oxygen in the blood, P_aO_2). The following changes are observed:

- **Hyperventilation**: Peripheral chemoreceptors in the carotid bodies sense low oxygen levels (P_aO_2) and stimulate increased breathing to take in more oxygen.

- **Respiratory alkalosis**: Hyperventilation causes excessive loss of CO_2, and the lowered P_aCO_2 leads to an increase in blood pH (respiratory alkalosis).
- **Acclimatization**: Over time, the body adapts by increasing red blood cell production (increased erythropoietin- EPO- production), a right shift of the oxyhemoglobin curve due to increased 2,3, diphosphoglycerate production (2,3 DPG), and increased blood volume and pulmonary capillary density increasing surface area for diffusion. These changes lead to improved oxygen delivery to tissues. Additionally, the pH of the blood CSF approaches normal (kidney metabolic response), and the peripheral receptors to hypoxia decrease.

Pathology: High-Altitude Pulmonary Edema (HAPE) is a life-threatening condition where fluid accumulates in the lungs during ascent to high altitudes. At high elevations, the decreased oxygen levels (hypoxia) cause generalized hypoxic vasoconstriction of the pulmonary blood vessels. This vasoconstriction increases pulmonary vascular resistance and arterial pressure. As a result of this increased hydrostatic pressure, fluid leaks from the pulmonary capillaries into the alveoli, leading to pulmonary edema. The accumulation of fluid impairs gas exchange, causing hypoxemia and respiratory distress.

Pathology: High-Altitude Cerebral Edema (HACE) is a severe condition involving swelling of the brain due to fluid accumulation when ascending above 2,500 meters (8,200 feet). In response to hypoxia at high altitudes, cerebral blood vessels vasodilate to increase blood flow to the brain and compensate for low oxygen levels. This vasodilation increases capillary permeability, allowing fluid to leak into brain tissue. The resulting cerebral edema leads to increased intracranial pressure, which can impair brain function and cause neurological symptoms.

Hypoxic vasoconstriction increases pulmonary vascular resistance and arterial pressure, leading to fluid leakage.

Respiratory responses to Exercise

Exercise increases the body's demand for oxygen and produces more CO_2, requiring adjustments in ventilation. The following changes are observed:

- **Increased ventilation rate, and cardiac output**: Ventilation rises in proportion to the intensity of exercise to meet O_2 consumption and remove excess CO_2 produced. The cardiac output also increases to meet the demand to move gases, nutrients, and wastes between the lungs and tissues.
- **Feedforward response**: Before metabolic changes occur, the brain anticipates the need for increased oxygen and ramps up breathing as soon as exercise begins. This is a **neural mechanism** rather than a response to changes in blood gases.
- **Chemoreceptor feedback**: As exercise continues, central and peripheral chemoreceptors respond to rising CO_2 and falling pH, further increasing

ventilation to stabilize these levels. Overall there is no change in the *mean* P_aO_2, P_aCO_2 and pH due to this homeostasis.

- **Ventilation to Perfusion (V/Q) ratios:** The airflow and blood flow in different parts of the lungs (Ventilation to Perfusion ratio) become more evenly distributed. This is because of the increased ventilation and perfusion and a decrease in dead space (regions of the lung that are not perfused).

Respiratory response to Diving

In diving, especially when breathing compressed air underwater, unique changes in ventilation occur:

- **Increased pressure**: As divers descend into the water's depths, the pressure increases, and the gases in the lungs become more compressed. This increases the partial pressures of gases, increasing the amount of dissolved nitrogen in the blood. If the ascent and decompression are not slow and controlled, the dissolved nitrogen can form bubbles, leading to gas emboli in peripheral or pulmonary circulation (chokes) or muscles and joints, causing pain called the bends. This is why nitrogen is replaced with helium in SCUBA tanks for deep dives.
- **Hypercapnia risk**: If breathing is not adequately adjusted, CO_2 can build up, causing hypercapnia (high CO_2 levels).
- **Shallow water blackout**: Hyperventilating and then breath-holding during diving can lead to a dangerous drop in oxygen without the usual rise in CO_2 that triggers breathing, risking unconsciousness.

CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to GB, why is she breathing abnormally?

GB's history and Biot breathing pattern (inspiratory pauses) are suggestive of an opioid overdose. This represses the CNS respiratory centers, causing a low respiratory rate, and can also lead to the observed abnormal breathing pattern.

Opioid intoxication can cause respiratory arrest and death, so it should be promptly reversed with an opioid receptor antagonist such as naloxone. This was done in the emergency department, and GB promptly became lucid, with normal breathing.

Summary

- Breathing is mostly involuntary, but people can exercise some voluntary control over their breathing.
- Chemoreceptors and mechanoreceptors contribute to the ventilatory responses.
- The dorsal respiratory group (DRG) in the medulla is responsible for inspiration and setting the respiratory rhythm, receiving signals from the glossopharyngeal and vagus nerves.
- The ventral respiratory group (VRG) in the medulla activates accessory respiratory muscles during active inspiration and expiration.
- The pneumotaxic center (PNC) is located in the upper pons; it acts to inhibit inspiration to regulate inspiratory volume and respiratory rate.
- The apneustic center (APC) is located in the lower pons; it acts to stimulate inspiration and can cause deep, gasping inspirations with occasional expirations.
- The carotid body and aortic chemoreceptors fire in response to decreases in blood partial pressure of oxygen (PO_2), as well as increases in blood

partial pressure of carbon dioxide (PCO_2) levels and reduced pH.

- The central (medullary) chemoreceptor in the brain senses decreases in cerebrospinal fluid (CSF) pH to increase minute ventilation.
- The mechanoreceptors that control medullary ventilation are pulmonary stretch receptors, irritant receptors, and J receptors.
- Hering-Breuer reflex results from mechanoreceptors sensing overdistention of the lungs and thus causes a decrease in respiratory rate.
- The different breathing patterns relate to different diseases or conditions.
- Hypoxic respiratory drive occurs when low oxygen levels, rather than high CO_2 levels, stimulate breathing. In chronic hypercapnia (e.g., COPD and emphysema), patients rely more on peripheral chemoreceptors that detet low O_2, for breathing. Excessive supplemental oxygen can lead to respiratory failure, so it should be titrated to maintain 88-92% oxygen saturation.
- Ventilation can adapt to different conditions such as altitudes, exercise, and diving.

Review Questions

1. In a person with no lung disease, which of the following would be the most likely effect of overdistention of the lungs?

A. Pulmonary stretch receptors activation leading to decreased respiratory rate

B. Irritant receptors activation causing an increase in respiratory rate

C. J receptor activation resulting in shallow, rapid breathing

D. Muscle mechanoreceptor activation leading to increased respiratory effort

Explanation (requires correct answer)

2. Which of the following structures acts to inhibit inspiration to regulate inspiratory volume and respiratory rate?

A. Apneustic center

B. Dorsal respiratory group

C. Medulla

D. Pneumotaxic center

E. Ventral respiratory group

Explanation (requires correct answer)

3. Which of the following is an efferent connection of the dorsal respiratory group?

A. Glossopharyngeal nerve

B. Laryngeal nerve

C. Phrenic nerve

D. Sympathetic chain

E. Vagus nerve

Explanation **(requires correct answer)**

4. A 74-year-old patient presents to the emergency department and is found to have a stroke involving his medulla. Which of the following abnormal breathing patterns is most likely?

A. Biot breathing

B. Cheyne-Stokes breathing

C. Kussmaul breathing

D. Obstructive sleep apnea

Explanation **(requires correct answer)**

5. A 68-year-old patient with COPD is admitted to the hospital with shortness of breath. High-flow supplemental oxygen is administered, and shortly after, the patient's respiratory rate decreases significantly. Which of the following best explains this response?

A. Central chemoreceptors are overstimulated by high CO_2 levels.

B. Peripheral chemoreceptors are no longer detecting hypoxemia.

C. Respiratory muscles are fatigued due to increased work of breathing.

D. Hypoxic drive has been suppressed by elevated PaO_2 from supplemental oxygen.

E. Baroreceptors have responded to increased blood pressure from oxygen therapy.

Explanation (requires correct answer)

6. Which of the following best explains the body's respiratory adaptation to intense exercise?

A. Chemoreceptor activation due to elevated arteriolar pH levels

B. Reduced cardiac output to maintain PaO_2 levels

C. Anticipatory feedforward mechanism increasing ventilation before metabolic changes occur

D. Increased $PaCO_2$ triggering hyperventilation

E. Increased PaO_2 due to enhanced oxygen uptake

Explanation **(requires correct answer)**

How would you rate this Brick?

References

- Webster, L. R., & Karan, S. (2020). The physiology and maintenance of respiration: a narrative review. *Pain and Therapy*, *9*(2), 467–486. https://doi.org/10.1007/s40122-020-00203-2
- Richerson G.B. and Boron W.F (2017). Chapter 32 Control of Ventilation in Medical Physiology 3rd ed. Editors Boron and Boulpaep. 700-720.e1

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 Identify the regions in the central nervous system involved in control of breathing.

02 Describe the locations, roles of chemoreceptors and their importance in detecting changes in arterial partial pressure of oxygen (PO_2), carbon dioxide (PCO_2), and pH.

03 Describe pulmonary receptor reflexes that influence breathing frequency and tidal volume.

04 Describe the hypoxic respiratory drive in a COPD patient and changes with supplemental oxygen.

05 Describe pathological breathing patterns (Kussmaul, Cheyne-Stokes, Biot, and apneustic).

06 Explain adaptive ventilation changes with altitudes, exercise, and diving.

Objective Confidence: 0/6